

1. Why does flocking emerge?

Flocking emerges because simple things that boids do like avoiding neighbors matching speed and staying close all add up to create movements.

2. Why is no leader required?

No leader is required because nobody is in charge of the flock. Every bird just reacts to what's around it and the group figures things out on its own.

3. What happens when the perception radius becomes very small?

When the perception radius becomes very small the flock breaks apart. The boids cannot see far so they cannot fly together without crashing into each other and it becomes chaotic.

4. What happens when the perception radius becomes very large?

When the perception radius becomes very large the flock becomes like a block. If every bird is paying attention to every bird then the flock loses its natural movement.

5. What causes a flock to fragment?

A flock fragments when the boids start moving from each other too much or when there are obstacles in their way or when predators show up and the boids try to escape.

6. How is Boids similar to Cellular Automata?

Boids is similar to Cellular Automata because they both use rules to create complex patterns.

7. How is Boids different from Neural Cellular Automata?

Boids is different from Neural Cellular Automata because Boids uses fixed rules that are set in advance. Neural Cellular Automata uses networks to learn its own rules.

8. What examples of behavior exist in the real world?

Examples of behavior, in the real world include ant colonies building nests, boids forming flocks traffic jams happening and crowds organizing themselves.